

D877

General Biology II

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A teacher candidate uses a video filmed at an invented skate park showing a scraped knee healing over two weeks. The video has over a million views. Which type of cell division repairs the skin?

- A. Mitosis, producing identical diploid cells
- B. Meiosis, because healing requires genetic variety
- C. Meiosis, producing haploid cells
- D. Binary fission

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A plant grown in a school greenhouse with automatic misters has a diploid number of 24. The seed packet said it would flower in 60 days. How many chromosomes are in each of its gametes?

- A. 48
- B. 24
- C. 12
- D. 6

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A gamete of an animal contains 9 chromosomes. How many chromosomes are in a skin cell that results from mitosis in this animal?

- A. 18
- B. 4.5
- C. 36
- D. 9

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

An insect species studied on a field trip to an invented nature preserve has a diploid number of 8. The preserve charges \$5 admission. Ignoring crossing over, how many different chromosome combinations can its gametes have?

- A. 4
- B. 8
- C. 16
- D. 256

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

Independent assortment alone gives an organism 32 possible gamete combinations. What is its diploid number?

- A. 10
- B. 16
- C. 32
- D. 5

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

Nerve cells and muscle cells from the same person look and work very differently. A biotech company advertises that it can sequence a genome in a day. Why are these cells different?

- A. Muscle cells contain more DNA because they are larger.
- B. Each cell type keeps only the genes it needs and loses the rest.
- C. Mutations during development gave them different DNA.
- D. They carry the same DNA but express different genes.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

Glucagon travels in the blood to every tissue, including the heart, which pumps about 5 liters per minute. Yet mainly liver cells respond by releasing glucose. Why?

- A. Other cells destroy glucagon before it can act.
- B. Hormones always act on all cells equally.
- C. Glucagon reaches only the liver.
- D. Liver cells have glucagon receptors that most other cells lack.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

In a simulation with a colorful touchscreen interface, a learner increases the strength of a stimulus to a sensory neuron from just above threshold to three times threshold. The simulation was built by an invented software startup. How does the neuron's response change?

- A. Each action potential becomes three times larger.
- B. It stops firing to protect itself.
- C. It fires at the same rate, since neurons always fire at one speed.
- D. It fires action potentials more often, each the same size.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A stimulus to a neuron is just below threshold. What happens?

- A. A full action potential fires, but slowly.
- B. The neuron fires at a higher frequency.
- C. No action potential fires.
- D. A small action potential fires.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A case study in a textbook with a glossy cover describes an invented adult who, after an injury, walks unsteadily and cannot smoothly touch a finger to the nose. Memory and speech are normal. Which brain region was most likely damaged?

- A. Medulla oblongata
- B. Cerebellum
- C. Hippocampus
- D. Frontal lobe

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

On a hot day at an outdoor track meet with 300 spectators, a runner's body temperature rises. Sweating increases and blood vessels near the skin widen until temperature returns toward 37 °C. What kind of control is this?

- A.** No feedback; sweating is a voluntary behavior
- B.** Negative feedback
- C.** Positive feedback, because it helps the body
- D.** Positive feedback, because sweating increases

Target word _____	The one fact that decides _____	Answer ____	[] sure [] unsure [] guess
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12. [TRANSFER — reversed stem / different variable]

Which process is an example of positive feedback?

- A.** Shivering when body temperature drops
- B.** Oxytocin increasing uterine contractions during childbirth, which release more oxytocin
- C.** Insulin lowering blood glucose after a meal
- D.** Breathing rate rising during exercise to remove CO₂

Target word _____	The one fact that decides _____	Answer ____	[] sure [] unsure [] guess
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STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A teacher candidate uses a video ~~filmed at an invented skate park~~ showing a scraped knee healing over two weeks. ~~The video has over a million views.~~ Which type of cell division repairs the skin?

Target word: division type for repair **Decides:** Repair needs identical diploid cells.

A	Correct	Skin repair replaces cells with copies.
B	Inverted operation	Repair needs identical cells, not varied ones.
C	Confusion set	Meiosis makes gametes, not skin cells.
D	Adjacent category	Prokaryotes divide this way; human cells do not.

2. Answer C Concept 2

A plant grown ~~in a school greenhouse with automatic misters~~ has a diploid number of 24. ~~The seed packet said it would flower in 60 days.~~ How many chromosomes are in each of its gametes?

Target word: gamete chromosome number **Decides:** Meiosis halves $2n$ once.

A	Inverted operation	Doubles instead of halving.
B	Dropped element	Leaves out the halving in meiosis.
C	Correct	$24 / 2 = 12$ (n).
D	Confusion set	Halves twice; meiosis II separates chromatids and does not halve the number again.

3. Answer A Concept 2 • transfer

A gamete of an animal contains 9 chromosomes. How many chromosomes are in a skin cell that results from mitosis in this animal?

Target word: solve for body-cell number **Decides:** Body cells are diploid: 2×9 .

A	Correct	$n = 9$, so $2n = 18$; mitosis keeps $2n$.
B	Inverted operation	Halves instead of doubling.
C	Confusion set	Doubles twice as if mitosis doubled the count.
D	Wrong stem number	That is the gamete number.

4. Answer C Concept 3

An insect species studied ~~on a field trip to an invented nature preserve~~ has a diploid number of 8. ~~The preserve charges \$5 admission.~~ Ignoring crossing over, how many different chromosome combinations can its gametes have?

Target word: combinations from independent assortment **Decides:** $2n = 8$ means $n = 4$ pairs; 2^4 .

A	Dropped element	Gives the number of pairs, not combinations.
B	Inverted operation	Multiplies 2×4 instead of raising 2 to the 4th.
C	Correct	$2^4 = 16$.
D	Confusion set	Uses 2^8 , the diploid number, instead of the number of pairs.

5. Answer A Concept 3 · transfer

Independent assortment alone gives an organism 32 possible gamete combinations. What is its diploid number?

Target word: solve backward for 2n **Decides:** $2^n = 32 \rightarrow n = 5$; $2n = 10$.

A	Correct	$n = 5$ pairs, so $2n = 10$.
B	Inverted operation	Divides 32 by 2 instead of taking the exponent.
C	Wrong stem number	Repeats the number of combinations.
D	Dropped element	That is n, not the diploid number.

6. Answer D Concept 4

Nerve cells and muscle cells from the same person look and work very differently. ~~A biotech company advertises that it can sequence a genome in a day.~~ Why are these cells different?

Target word: basis of specialization **Decides:** Same genome, different genes expressed.

A	Inverted operation	Cell size does not change the genome.
B	Confusion set	Nearly all body cells keep the full genome.
C	Adjacent category	Specialization is regulated, not caused by random mutation.
D	Correct	Gene regulation, not gene content, creates cell types.

7. Answer D Concept 5

Glucagon travels in the blood to every tissue, including the heart, ~~which pumps about 5 liters per minute.~~ Yet mainly liver cells respond by releasing glucose. Why?

Target word: why only some cells respond **Decides:** Only target cells have the receptor.

A	Confusion set	Non-target cells simply lack the receptor.
B	Absolute language	Hormones act only on cells with matching receptors.
C	Adjacent category	It circulates everywhere; the stem says so.
D	Correct	Receptors decide which cells respond to a signal.

8. Answer D Concept 6

In a simulation ~~with a colorful touchscreen interface,~~ a learner increases the strength of a stimulus to a sensory neuron from just above threshold to three times threshold. ~~The simulation was built by an invented software startup.~~ How does the neuron's response change?

Target word: response to a stronger stimulus **Decides:** Action potentials are all-or-none.

A	Inverted operation	Spike size is fixed.
B	Adjacent category	A strong stimulus increases firing.
C	Absolute language	Firing frequency changes with stimulus strength.
D	Correct	Strength is coded by frequency.

9. Answer C Concept 6 · transfer

A stimulus to a neuron is just below threshold. What happens?

Target word: response below threshold **Decides:** Below threshold, no action potential fires.

A	Adjacent category	Nothing fires below threshold.
B	Inverted operation	Higher frequency comes from stronger, not weaker, stimuli.
C	Correct	All-or-none: threshold must be reached.
D	Confusion set	There are no partial action potentials.

10. Answer B Concept 7

A case study ~~in a textbook with a glossy cover~~ describes an invented adult who, after an injury, walks unsteadily and cannot smoothly touch a finger to the nose. Memory and speech are normal. Which brain region was most likely damaged?

Target word: brain region from symptoms **Decides:** Balance and coordination point to the cerebellum.

A	Confusion set	Controls breathing and heart rate.
B	Correct	Coordinates balance and fine movement.
C	Adjacent category	Forms new memories, which are normal here.
D	Dropped element	Plans movement but coordination errors point to the cerebellum.

11. Answer B Concept 8

On a hot day ~~at an outdoor track meet with 300 spectators~~, a runner's body temperature rises. Sweating increases and blood vessels near the skin widen until temperature returns toward 37 °C. What kind of control is this?

Target word: feedback type **Decides:** The response reverses the change.

A	Adjacent category	Sweating is an automatic response.
B	Correct	It counteracts the rise and returns to the set point.
C	Confusion set	"Positive" means amplifying, not helpful.
D	Inverted operation	The effector output rises, but its effect reverses the change.

12. Answer B Concept 8 · transfer

Which process is an example of positive feedback?

Target word: identify positive feedback **Decides:** Positive feedback amplifies until an endpoint.

A	Confusion set	Restores the set point; negative feedback.
B	Correct	Each round amplifies the next until birth.
C	Inverted operation	Reverses a rise; negative feedback.
D	Adjacent category	Corrects rising CO ₂ ; negative feedback.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	8	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	10	
Absolute language	Always, never, guarantee — where judgment applies.	2	
Confusion set	Every option is real; the test is exact role-matching.	10	
Camouflage	Answers a stem fact that was never the question.	0	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Mitosis vs. meiosis outcomes	Mitosis gives two genetically identical diploid cells for growth and repair; meiosis gives four genetically different haploid gametes.	Linking meiosis to healing or growth.	The purpose of the division: repair vs. reproduction.
2	Chromosome number through meiosis	Meiosis halves the chromosome number once ($2n \rightarrow n$); fertilization restores $2n$; mitosis keeps the number the same.	Halving twice (once per meiotic division) or doubling.	A diploid number and a question about gametes or body cells.
3	Independent assortment	With n chromosome pairs, independent assortment alone gives 2^n possible gamete combinations.	Multiplying $2 \times n$ or using $2n$ in the exponent.	"How many different combinations of chromosomes...?"
4	Differentiation is gene expression	Specialized cells of one organism carry the same DNA; they differ in which genes are turned on.	Thinking cells lose or gain genes when they specialize.	Two cell types from the same organism compared.
5	Receptor specificity in signaling	A signal affects only cells that have its receptor; second messengers inside the cell amplify the response.	Thinking a hormone acts only where it is delivered.	A hormone in the blood that affects only some tissues.
6	All-or-none action potentials	An action potential either fires fully or not at all; a stronger stimulus raises the firing frequency, not the size of each spike.	Thinking a stronger stimulus makes a bigger action potential.	Stimulus strength changes and the question asks about the neuron's response.
7	Brain region functions	Cerebellum = balance and coordination; medulla = breathing and heart rate; hippocampus = forming new memories; frontal lobe = planning and decisions; occipital lobe = vision.	Mixing cerebellum with cerebrum or medulla.	A symptom or task linked to a brain area.
8	Negative vs. positive feedback	Negative feedback reverses a change to hold a set point; positive feedback amplifies a change until an endpoint is reached.	Calling any feedback that "helps" the body positive.	A body response to a change from normal.

Course scope (public description): This General Biology II course is designed to prepare learners for teaching biology at the high school level by providing an in-depth understanding of cellular structures, the processes of mitosis and meiosis, and the mechanisms of communication and information processing within organisms. Students will explore the complexities of cell differentiation, the roles of different brain regions, and how organisms respond to environmental stimuli through interactive activities and simulations. The course emphasizes the integration of foundational concepts with real-world applications, helping future educators build a comprehensive knowledge base. By the end of the course, students will be well-equipped to teach key biological processes and understand their significance in the growth, development, and survival of living organisms.